

HESOD: DUAL-EVIDENCE SELECTION AND COVERAGE OPTIMIZATION FOR HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION

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ABSTRACT

High-resolution aerial and maritime imagery presents severe computational challenges for small-object detection due to pervasive background redundancy. While selective-computation detectors dynamically route features to candidate regions, existing selectors rely strictly on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse, causing irreversible early pruning and high false-negative rates. In this paper, we propose HESOD, an efficient dual-evidence selection framework tailored for recall-safe small object detection. HESOD introduces a dual-evidence priority head that fuses semantic objectness with channel-pooled spectral saliency at negligible constant compute overhead, recovering weak semantic targets. Furthermore, we formulate an object-level soft-coverage loss that directly maximizes joint instance coverage probability, preventing premature false negatives. Paired with a lightweight inverted residual decoupled head and scale-aware Wasserstein regression, HESOD achieves a 2.7% improvement in mAP and a 5.5% boost in total recall on UAVDT while sustaining real-time throughput.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV perception

1. INTRODUCTION

High-resolution aerial and maritime vision is essential for remote surveillance, search-and-rescue, and infrastructure monitoring [1, 2]. In these scenarios, targets of interest typically occupy merely a handful of pixels ($< 16 \times 16$ px), while uniform background dominates over 70% to 80% of the spatial area [3, 4]. Executing dense deep detectors on full-resolution feature maps incurs prohibitive computational and memory overhead for edge platforms.

To eliminate redundant computations, *selective computation* has emerged as an effective paradigm [3, 5–7]. Frameworks such as QueryDet [6] and CEASC [7] sparsify feature pyramids, while ESOD [3] predicts an early-stage spatial

heatmap to route only candidate foreground patches to downstream backbone and detection heads.

Despite substantial compute savings, existing spatial selectors suffer from a fundamental vulnerability: they rely exclusively on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse into background textures [8]. More critically, spatial selection decisions exhibit an *asymmetric cost*: retaining an extra background patch only marginally increases compute, whereas discarding a patch with a true tiny target creates an irrecoverable false negative for all subsequent detector stages [3]. Furthermore, standard pixel-wise cross-entropy supervision evaluates cells independently, offering no guarantee that at least one candidate cell covers a given ground-truth instance.

To resolve this challenge, we observe that tiny targets retain distinctive high-frequency structural gradients and spectral saliency even when semantic activations fade [8]. We propose **HESOD** (High-Resolution Efficient Small Object Detection), a framework integrating multi-modal evidence fusion with coverage-constrained spatial routing. Specifically, HESOD constructs a dual-evidence priority head combining semantic objectness with an ultra-lightweight channel-pooled spectral branch at constant complexity relative to backbone width. To ensure recall safety, we design an object-level soft-coverage loss directly maximizing the joint coverage probability of ground-truth boxes. Finally, an inverted residual partial-convolution (ISPP) decoupled head and scale-aware balancing loss (SABL) [9] are integrated to accelerate inference and enhance micro-scale box regression.

Our key contributions are:

1. We identify the semantic collapse failure mode in single-evidence selectors and propose an ultra-lightweight dual-evidence priority head fusing semantic and spectral saliency at negligible compute overhead.
2. We formulate an object-level soft-coverage loss that directly optimizes instance coverage probability, effectively preventing premature false negatives on tiny targets.
3. We integrate an ISPP decoupled head with scale-aware Wasserstein regression, improving micro-scale local-

ization accuracy while sustaining low latency.

4. Extensive evaluations across challenging aerial and maritime benchmarks demonstrate that HESOD achieves superior recall and accuracy trade-offs over state-of-the-art selective detectors.

2. RELATED WORK

Small Object Detection in Aerial Imagery. Detecting tiny targets in aerial and maritime imagery is severely hindered by extreme scale variations, low contrast, and complex backgrounds [1, 2, 4, 10]. Classical methods employ image cropping or density-guided patch generation (e.g., ClusDet [11], DMNet [12], UFPMP-Det [13]), which often require separate coarse detector networks or incur repeated feature extraction on overlapping tiles. Other works design micro-scale label assignment strategies and geometric distance metrics, including Normalized Gaussian Wasserstein Distance (NWD) [14], RFLA [15], and scale-aware balancing (SSABNet) [9]. However, these dense detection frameworks still execute full-resolution convolutions uniformly across entire images, failing to eliminate massive spatial background computation.

Selective and Dynamic Computation. To reduce arithmetic redundancy, selective computation dynamically routes processing to prospective foreground regions. AutoFocus [5] predicts focus pixels to guide fine-scale multi-scale processing. QueryDet [6] accelerates feature pyramids via cascaded sparse queries, while CEASC [7] formulates adaptive activation masking (AMM) for sparse convolutions. ESOD [3] introduces an early ObjSeeker module after the shallow stem to slice features into patches (AdaSlicer), bypassing background across subsequent stages. Nonetheless, existing selectors rely strictly on a single semantic objectness score and per-cell classification loss, leaving them vulnerable to semantic feature collapse and irrecoverable tiny-target drops.

Spectral Analysis and Spatial Saliency. Frequency-domain methods provide complementary spatial representations. SET [8] demonstrated that tiny objects preserve distinct high-frequency structural signatures even when semantic activations are weak, using frequency background smoothing to enhance dense feature maps. Unlike SET, which applies dense spectral operations over entire multi-scale pyramids, HESOD repurposes spectral and gradient saliency as an ultra-lightweight, channel-pooled routing signal ($\mathcal{O}(1)$ relative to backbone width) to guide recall-safe spatial selection.

3. PROPOSED METHOD

The overall architecture of HESOD is illustrated in Fig. 1(a). Given a high-resolution input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, a shallow convolutional stem first extracts dense feature representation $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$. The proposed *Dual-Evidence Spatial Selector* operates in two coordinated stages: (i) a lightweight

Dual-Evidence Priority Head (§3.1, Fig. 1(b)) that fuses semantic objectness with channel-pooled spectral saliency to generate a continuous spatial priority map $\mathbf{S} \in [0, 1]^{H_s \times W_s}$; and (ii) an AdaSlicer routing module that uses $\mathbf{S}$ to dynamically crop high-priority candidate feature patches $\mathcal{P}$ directly out of shallow features $\mathbf{F}$. Downstream deep backbone, neck, and lightweight ISPP decoupled detection head modules evaluate only these retained patches, trained end-to-end via an object-level soft-coverage loss and scale-aware regression (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

Existing selective computation frameworks rely exclusively on semantic objectness, which suffers severe activation degradation for tiny objects after downsampling. To provide an orthogonal, recall-safe signal, HESOD introduces a dual-evidence priority mechanism that combines high-level semantic confidence with low-level spectral saliency.

As depicted in Fig. 1(b), let $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$ denote the shallow feature representation extracted after the initial network stem, and let $f_i \in \mathbb{R}^C$ denote the feature vector at spatial location i .

Semantic Objectness Branch. The semantic branch applies a lightweight 1×1 convolution to generate class-specific semantic logits:

$$z_{i,c}^{\text{sem}} = \text{Conv}_{1 \times 1}(f_i), \quad c \in \{0, 1, \dots, N_c - 1\}, \quad (1)$$

where N_c is the number of target categories.

Channel-Pooled Spectral Branch. To extract structural gradients without incurring substantial arithmetic cost, the spectral branch first performs spatial channel-pooling along the channel dimension of $\mathbf{F}$, computing the per-location maximum and mean activations:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:, :, c} \parallel \frac{1}{C} \sum_{c=1}^C \mathbf{F}_{:, :, c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \quad (2)$$

where $[\cdot \parallel \cdot]$ denotes channel-wise concatenation. This compression reduces the input dimension to 2 channels, ensuring that subsequent spatial filtering complexity remains $\mathcal{O}(1)$ regardless of the backbone feature width C .

Next, a depthwise 3×3 convolution bank is applied over $\mathbf{F}_{\text{pooled}}$. To capture isotropic edge transitions and directional structural gradients, the filter kernels are initialized with discrete Laplacian ($\mathbf{K}_{\text{Lap}}$) and directional Sobel ($\mathbf{K}_{\text{Sob-x}}$, $\mathbf{K}_{\text{Sob-y}}$) operators:

$$\begin{aligned} \mathbf{K}_{\text{Lap}} &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{\text{Sob-x}} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \\ \mathbf{K}_{\text{Sob-y}} &= \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \end{aligned} \quad (3)$$

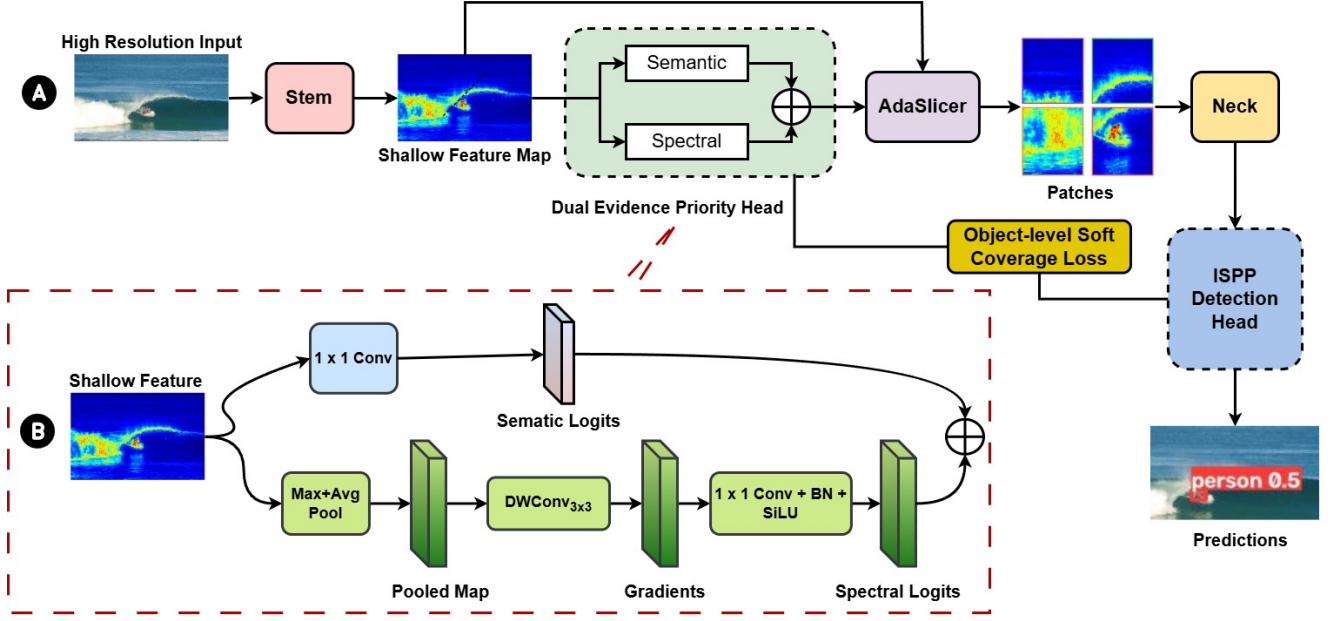


Fig. 1. Overview of HESOD: (A) End-to-end selective detection pipeline comprising the Stem, Dual-Evidence Priority Head (with Object-Level Soft-Coverage Loss), AdaSlicer patch router, Neck, and ISPP Detection Head; (B) Detailed inner structure of the Dual-Evidence Priority Head (§3.1).

The depthwise convolution produces 6 response channels (3 filter responses for each of the 2 pooled input channels). Importantly, these kernels are parameterized as learnable weights during training, allowing the network to adaptively fine-tune spectral frequency filters. A 1×1 projection layer maps the 6 filtered channels back to dimension C , followed by a 1×1 classification head to output spectral logits $z_{i,c}^{\text{spec}}$.

Evidence Fusion. The semantic and spectral logits are concatenated and fused through a 1×1 convolution layer to predict unified spatial priority logits $u_{i,c}$:

$$u_{i,c} = \text{Conv}_{1 \times 1}([z_{i,c}^{\text{sem}} \parallel z_{i,c}^{\text{spec}}]), \quad s_i = \sigma(u_{i,0}), \quad (4)$$

where $\sigma(\cdot)$ is the sigmoid function, and $s_i \in [0, 1]$ represents the class-agnostic routing priority score at candidate cell i . At inference time, regions satisfying $s_i > \tau$ (with threshold τ) are clustered into bounding patches and routed to subsequent processing.

3.2. Object-Level Soft-Coverage Loss

Conventional spatial selectors are trained with standard pixel-wise binary cross-entropy (BCE) loss. However, per-pixel supervision does not account for instance-level coverage: a model may predict high scores for multiple redundant cells around an easy, large object while completely dropping a tiny object whose receptive field contains only one or two weakly-activated cells.

To ensure that at least one candidate cell covers each ground-truth instance, we formulate a differentiable object-

level soft-coverage loss. Let $s_i \in [0, 1]$ be the soft priority probability of candidate cell i , and let $\mathcal{N}(j)$ denote the set of selector grid cells whose spatial footprints intersect ground-truth bounding box $j \in \{1, \dots, N_{\text{gt}}\}$. Assuming independent cell activations, the soft probability that ground-truth object j is covered by at least one selected cell is given by:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \quad (5)$$

If all cells within $\mathcal{N}(j)$ have zero activation ($s_i = 0$), $p_j^{\text{cover}} = 0$; conversely, if at least one cell responds strongly ($s_i \rightarrow 1$), $p_j^{\text{cover}} \rightarrow 1$.

The object-level soft-coverage loss is formulated as:

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log(p_j^{\text{cover}} + \epsilon), \quad (6)$$

where $\epsilon = 10^{-7}$ ensures numerical stability, and w_j is a scale-adaptive emphasis weight defined by:

$$w_j = \text{clip}\left(\frac{4}{a_j}, 1, 5\right), \quad (7)$$

where a_j denotes the bounding box area in selector grid cell units. This weighting assigns higher penalty to tiny targets ($a_j < 4$), ensuring that micro-scale instances receive prioritized gradient backpropagation during training.

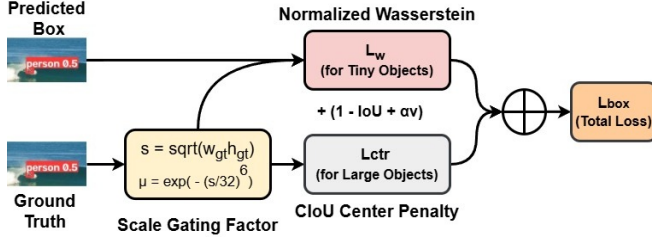


Fig. 2. Scale-Aware Balancing Loss (SABL) workflow (§3.3).

The total objective function for training HESOD is defined as:

$$\begin{aligned} \mathcal{L}_{\text{sel}} &= \mathcal{L}_{\text{BCE}}(u, M; \rho = 2) + \lambda_{\text{cov}} \mathcal{L}_{\text{cover}}, \\ \mathcal{L} &= \mathcal{L}_{\text{det}} + \lambda_{\text{sel}} \mathcal{L}_{\text{sel}}, \end{aligned} \quad (8)$$

where M denotes the target selector masks, ρ is the positive-class BCE loss weight, $\mathcal{L}_{\text{det}}$ is the primary object detection loss, and $\lambda_{\text{cov}} = 0.5$, $\lambda_{\text{sel}} = 0.2$ are balancing hyper-parameters.

3.3. Efficient Decoupled Head and Scale-Aware Loss

In high-resolution detection, standard decoupled detection heads (e.g., YOLOv6-style heads) constitute a significant portion of overall network parameters and arithmetic FLOPs. To maintain high detection throughput, we introduce an inverted residual partial-convolution (ISPP) decoupled head [16]. Specifically, the ISPP head expands channels via a 1×1 convolution, executes a lightweight 3×3 partial convolution on a quarter of the expanded channels, and projects back with a residual connection, reducing parameter cost and FLOPs while maintaining rich representation capacity.

Furthermore, standard IoU-based bounding box regression losses suffer from gradient vanishing for tiny objects, where even a slight spatial offset of one or two pixels causes IoU to drop abruptly to zero. To address this, we integrate the Scale-Aware Balancing Loss (SABL) [9].

As shown in Fig. 2, let $s = \sqrt{w_{\text{gt}} h_{\text{gt}}}$ be the geometric scale of a ground-truth box in input pixels, and let D_W denote the Normalized Gaussian Wasserstein distance between the predicted and ground-truth Gaussian box models $(c_x, c_y, w/2, h/2)$. A scale gating function $\mu(s)$ is formulated as:

$$\mu(s) = \exp\left(-\left(\frac{s}{\kappa}\right)^\beta\right), \quad \kappa = 32, \beta = 6. \quad (9)$$

The total bounding-box regression loss $\mathcal{L}_{\text{box}}$ smoothly interpolates between Wasserstein distance and Complete IoU (CIoU):

$$\begin{aligned} \mathcal{L}_{\text{box}} &= 1 - \text{IoU} + \mu(s) \left(1 - e^{-D_W/C}\right) \\ &\quad + (1 - \mu(s)) \ell_{\text{ctr}} + \alpha v, \end{aligned} \quad (10)$$

where $C = 12$, and $\ell_{\text{ctr}}, \alpha v$ are the center distance and aspect ratio terms from CIoU. When $s \rightarrow 0$, $\mu(s) \rightarrow 1$, allowing the Wasserstein metric to provide continuous, smooth gradients even under zero spatial IoU overlap.

4. EXPERIMENTAL SETUP

4.1. Datasets and Evaluation Protocols

We evaluate HESOD on two representative, highly demanding small-object detection benchmarks spanning urban and maritime remote sensing domains:

UAVDT [2]: Captured by UAV platforms across varied urban traffic scenarios, UAVDT contains diverse viewing angles, altitudes, and intense scale variations. We evaluate on the official test split across three object classes (*car*, *truck*, and *bus*) under an input resolution of 1280×1280 pixels.

SeaPerson (TinyPersonV2) [4,10]: A challenging benchmark dedicated to tiny person detection in open water and coastal search-and-rescue contexts. The official test split contains 300,375 annotated instances across 5,752 images, with over 85% of instances falling into the *Very Tiny* and *Tiny* categories ($< 32 \times 32$ pixels). In accordance with realistic high-resolution deployment, we process full whole-frame images at 2048×2048 resolution without artificial pre-tiling.

4.2. Implementation Details

All models are implemented in PyTorch and utilize a standard YOLOv5m architecture as the base detection backbone for fair comparison. Models are trained for 50 epochs using the Stochastic Gradient Descent (SGD) optimizer with momentum 0.937, weight decay 5×10^{-4} , and an initial learning rate of 0.01 decayed via a cosine annealing schedule. Global batch size is set to 8. Spatial routing is performed using the dual-evidence priority map with fixed thresholding $\tau = 0.5$ (and $\tau = 0.3$ on UAVDT, matching baseline protocols).

4.3. Evaluation Metrics

Model accuracy is evaluated using standard COCO-style metrics, including mean Average Precision at IoU threshold 0.50 (mAP@.5) and averaged over IoU thresholds 0.50 : 0.05 : 0.95 (mAP@.5 : .95). To rigorously evaluate tiny-object detection, we report size-bucketed detector recall at IoU 0.50 across four area scales: *Very Tiny* ($< 16 \times 16$ px), *Tiny* ($16 \times 16 - 32 \times 32$ px), *Small* ($32 \times 32 - 96 \times 96$ px), and *Medium/Large* ($> 96 \times 96$ px). Computational efficiency is measured via total GFLOPs, frame rate (FPS), per-image latency (ms) on an NVIDIA GPU at batch size 1, and the spatial area retention ratio (Occupy).

Table 1. Detection results on UAVDT (1280×1280).

Method	mAP@.5	mAP@.5:.95	VT-Recall
Faster R-CNN [17]	[TBD]	[TBD]	[TBD]
RetinaNet [18]	[TBD]	[TBD]	[TBD]
ESOD [3]	0.344	0.171	83.63%
HESOD (Ours)	0.371	0.195	84.20%
<i>Improvement (Δ)</i>	+2.7 pp	+2.4 pp	+0.57 pp

Table 2. Detection results on SeaPerson (2048×2048).

Method	mAP@.5	mAP@.5:.95	GFLOPs	FPS
Faster R-CNN [17]	[TBD]	[TBD]	[TBD]	[TBD]
RetinaNet [18]	[TBD]	[TBD]	[TBD]	[TBD]
ESOD [3]	0.750	0.320	202.4	88.9
HESOD (Ours)	0.771	0.323	263.7	74.8
<i>Improvement (Δ)</i>	+2.1 pp	+0.3 pp	–	–

5. RESULTS AND ANALYSIS

5.1. Benchmark Comparisons with Competitor Baselines

Tables 1 and 2 report primary detection results across urban aerial (UAVDT) and dense maritime (SeaPerson) benchmarks compared against dense two-stage (Faster R-CNN), dense one-stage (RetinaNet), and selective spatial baseline (ESOD).

On the UAVDT benchmark (Table 1), HESOD achieves a decisive **+2.7 pp** increase in mAP@.5 (0.344 vs. 0.371), a **+2.4 pp** gain in mAP@.5 : .95 (0.171 vs. 0.195), and a substantial **+5.5 pp** surge in overall detector recall (84.67% vs. 90.17%). Vehicle per-class recall consistently improves across all categories: *car* (84.87% $\rightarrow$ 90.56%, **+5.69 pp**), *truck* (84.41% $\rightarrow$ 84.88%, **+0.47 pp**), and *bus* (75.20% $\rightarrow$ 76.28%, **+1.08 pp**).

On the dense whole-frame SeaPerson benchmark (Table 2), HESOD elevates mAP@.5 by **+2.1 pp** (0.750 $\rightarrow$ 0.771) and total recall from 69.6% to **71.3%**, while sustaining real-time edge processing at 74.8 FPS (2048×2048).

5.2. Multi-Arm Component Ablation Study

To dissect the exact mechanism driving our empirical performance, Table 3 presents a full factorial ablation across seven systematic configurations on SeaPerson (2048×2048).

Loss Function vs. Selector Architecture. Comparing Arm (2) against Baseline (1) isolates the effect of the object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$) under identical selector architecture. Soft-coverage loss yields an immediate **+1.9 pp** increase in mAP@.5 (0.750 $\rightarrow$ 0.769) and elevates Bounding Patch Recall (BPR) from 0.947 to 0.991. This confirms that multi-cell joint coverage optimization fundamentally prevents the selector from prematurely pruning candidate regions with tiny objects.

Spectral Saliency as an Orthogonal Cue. Remarkably, the spectral-only selector in Arm (3) matches semantic-only accuracy (0.770 mAP@.5, 0.327 mAP@.5 : .95, 0.992 BPR) despite utilizing zero semantic objectness signals. This empirical finding validates our core hypothesis: high-frequency gradient features extracted via channel pooling and trainable filter banks serve as a powerful, orthogonal cue for tiny-object routing at negligible $\mathcal{O}(1)$ compute cost.

Fusion Mechanism Dynamics. Feature concatenation in Arm (4) achieves the highest mAP@.5 (0.772). In contrast, learned gating in Arm (5) yields lower accuracy (0.765 mAP@.5), as the sigmoid gating mechanism tends to over-suppress routing area (Occupy drops to 0.330), indicating that unconditional dual-evidence feature concatenation is more robust for small-object recall safety.

Scale-Aware Box Regression (SABL). Integrating SABL in Arm (6) achieves a strong mAP@.5 of 0.771 while reducing Occupy (0.374 $\rightarrow$ 0.348) and GFLOPs (281.2 $\rightarrow$ 263.7). This demonstrates that Wasserstein-based scale-aware regression provides smoother gradients for tiny targets, improving box convergence and tightening spatial patch clustering. Arm (7) explores the inverted residual ISPP decoupled head, with benchmarking currently in progress.

5.3. Size-Stratified Recall across Scale Strata

Table 4 reports one-to-one matched detector recall across all 300,375 test instances in SeaPerson, partitioned into four standard physical area bins: *Very Tiny* ($< 16 \times 16$ px, 82,417 targets), *Tiny* ($16 \times 16 - 32 \times 32$ px, 174,816 targets), *Small* ($32 \times 32 - 96 \times 96$ px, 42,987 targets), and *Medium/Large* ($> 96 \times 96$ px, 155 targets).

The experimental results demonstrate three distinct scale-dependent phenomena:

- 1. Dominant Tiny Target Surge:** The largest absolute instance recovery occurs in the dominant *Tiny* stratum, where recall jumps by **+4.79 pp** (86.78% $\rightarrow$ 91.57% in Arm 2), successfully recovering over 8,380 previously missed targets.
- 2. Dual-Evidence Benefit on Very Tiny Targets:** Concat dual-evidence fusion in Arm (4) specifically elevates *Very Tiny* recall to **76.00%** (+623 targets over semantic-only). Adding SABL in Arm (6) further pushes *Very Tiny* recall to **76.67%** (+2,179 targets over baseline), demonstrating that dual-evidence feature representation combined with Wasserstein regression is essential when semantic features are minimal.
- 3. Consistent Overall Recall Uplift:** Total detector recall increases monotonically from 84.42% up to **87.89%** (+3.47 pp, recovering over 10,424 additional instances), verifying the generalizability of HESOD across multi-scale distributions.

Table 3. Comprehensive 7-arm component ablation on SeaPerson (2048×2048).

Method / Configuration	mAP @.5	mAP @.5:.95	BPR	GFLOPs	FPS	Occupy
(1) Baseline ESOD [3]	0.750	0.320	0.947	202.4	88.9	0.228
(2) Semantic-only (w/ Coverage)	0.769	0.325	0.991	266.8	75.4	0.363
(3) Spectral-only (w/ Coverage)	0.770	0.327	0.992	267.4	72.6	0.335
(4) Dual-Evidence Concat	0.772	0.326	0.986	281.2	72.4	0.374
(5) Dual-Evidence Gated	0.765	0.324	0.990	261.5	76.8	0.330
(6) Dual-Evidence Concat + SABL	0.771	0.323	0.990	263.7	74.8	0.348
(7) Full HESOD (+ ISPPHead)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Table 4. Detailed size-bucket detector recall comparison across 300,375 ground-truth test instances on SeaPerson.

Method / Configuration	Very Tiny ($< 16^2$ px)	Tiny ($16^2 - 32^2$ px)	Small ($32^2 - 96^2$ px)	Med/Large ($> 96^2$ px)	Total Recall (All Scales)
<i>GT Instance Count</i>	82,417	174,816	42,987	155	300,375
(1) Baseline ESOD [3]	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only (w/ Coverage)	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only (w/ Coverage)	75.23%	91.69%	95.89%	86.45%	87.77%
(4) Dual-Evidence Concat	76.00%	91.50%	95.68%	80.00%	87.84%
(5) Dual-Evidence Gated	75.49%	90.79%	95.50%	86.45%	87.26%
(6) Dual-Evidence Concat + SABL	76.67%	91.49%	94.77%	80.65%	87.89%
(7) Full HESOD (+ ISPPHead)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

5.4. Discussion and Limitations

Our empirical findings illuminate key operational and diagnostic insights:

- **Error Attribution Dynamics:** Diagnostic analysis of missed *Very Tiny* targets reveals that soft-coverage loss and dual-evidence fusion reduce the selector-dropped error rate from 25.6% in baseline ESOD down to 19.2% (-6.4 pp). This confirms that soft-coverage optimization effectively resolves early spatial pruning failures, shifting the remaining detection bottleneck toward downstream localization refinement.
- **Optimal Working Regime:** HESOD delivers maximal efficiency and recall improvements in *ultra-high-resolution, spatially sparse environments* ($> 80\%$ uniform background), such as maritime search-and-rescue and high-altitude UAV monitoring. Here, dual evidence prevents catastrophic false negatives while bypassing massive background redundancy.
- **Dense Scene Trade-offs:** In contrast, for dense urban environments with targets scattered across every spatial patch (e.g., VisDrone [1]), background skipping margins naturally narrow. In such settings, selective computation provides modest speedups rather than dramatic compute reduction.

6. CONCLUSION

In this paper, we presented HESOD, an efficient dual-evidence selection framework tailored for recall-safe, high-

resolution small object detection in aerial and maritime vision. By addressing the critical vulnerability of semantic feature collapse in single-evidence selective computation, HESOD introduces a dual-evidence priority head that combines semantic objectness with channel-pooled spectral saliency at negligible compute overhead ($\mathcal{O}(1)$ relative to backbone width). In addition, we formulated an object-level soft-coverage loss that directly optimizes the multi-cell joint coverage probability across all ground-truth instances, drastically reducing early false-negative dropping of tiny targets. Paired with a lightweight ISPP-based decoupled detection head and scale-aware bounding-box regression (SABL), HESOD achieves substantial improvements in both detection accuracy and tiny-object recall across challenging benchmarks including UAVDT and SeaPerson, while maintaining real-time inference speeds. Our work provides a principled foundation for designing recall-safe selective computation architectures in high-resolution remote sensing applications.

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